AareML

Predicting River Water Quality in Swiss Catchments

CAS in Advanced Machine Learning
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April 2026 · Deadline: 15 June 2026
Report version: 1.10 · Last updated: 25 Apr 2026, 18:38

Abstract. AareML applies a sequence-to-sequence LSTM, modelled on the LakeBeD-US benchmark (McAfee et al., 2025), to predict dissolved oxygen (DO) and water temperature at 14-day horizons across 86 Swiss gauging stations (115 CAMELS-CH catchments, 86 with daily sensor records) from the CAMELS-CH-Chem dataset (Nascimento et al., 2025). Starting from three statistical baselines, we train a single-site LSTM (Optuna-tuned over 75 trials; default config achieves DO RMSE = 0.308 mg/L, Optuna best = 0.319 mg/L) and evaluate its zero-shot transfer to 12 Swiss gauges (mean DO RMSE = 0.427 mg/L, 3.3× better than LakeBeD-US reference; per-gauge retraining achieves 0.388 mg/L). A Wilcoxon signed-rank test across 11 gauges confirms a statistically significant improvement ($p=0.005$) for zero-shot transfer over Ridge regression. GradientSHAP attribution identifies temperature[t−1] as the dominant driver (mean $|\text{SHAP}|=0.644$), ahead of DO itself, with the LSTM exhibiting an effective memory of 3–4 days despite a 21-day lookback. Baseline DO RMSE on gauge 2473 ranges from 0.30–0.34 mg/L, already well below the LakeBeD-US lake reference of 1.40 mg/L, suggesting rivers are more predictable than lakes under the same LSTM architecture. Zero-shot temperature transfer across 15 gauges achieves mean RMSE of 2.59°C (NSE=0.730), with low-elevation gauges approaching single-site performance. Zero-shot application to 4 US rivers achieves mean RMSE of 1.527 mg/L, with the Willamette River (Oregon) approaching the lake benchmark at 1.007 mg/L. GradientSHAP and cross-ecosystem results are fully presented in this report.

Dataset	Method	Targets	Metric	Status
CAMELS-CH-Chem	Seq2Seq LSTM	DO · Temp	RMSE · KGE	In progress

1. Introduction

Dissolved oxygen (DO) is a primary indicator of aquatic health: concentrations below 5 mg/L stress fish populations, while hypoxic events (below 2 mg/L) cause mass mortality. Water temperature is equally critical: sustained temperatures above 18°C stress cold-water species such as trout and salmon, above 21°C disrupt migration and increase disease risk, and above 25°C are lethal for many native Alpine fish species. Temperature also controls DO solubility directly (cold water holds more oxygen), making it the dominant physical driver of DO dynamics — a relationship confirmed by the SHAP analysis in Section 5.4. In Swiss river networks, DO and temperature dynamics are driven by photosynthesis, reaeration, and upstream nutrient loads — a complex interplay that statistical models have historically struggled to capture at multi-day forecast horizons.

Recent work on lake water quality has demonstrated that sequence-to-sequence Long Short-Term Memory networks (LSTMs) can outperform traditional baselines substantially. McAfee et al. (2025) introduced LakeBeD-US, a benchmarking dataset for 21 US lakes with a seq2seq LSTM achieving a test RMSE of 1.40 mg/L for DO at a 14-day forecast horizon. Their architecture — 21-day lookback, teacher-forced decoder, masked MSE loss — provides a reproducible benchmark that we adapt here for river systems.

The CAMELS-CH-Chem dataset (Nascimento et al., 2025), published in early 2025, provides daily high-frequency sensor measurements of temperature, pH, electrical conductivity, and dissolved oxygen at 86 Swiss gauging stations from 1981 to 2020, complemented by 38-variable chemistry grab samples, land cover, and 115 catchment attribute files. To our knowledge, no published machine learning study has yet applied predictive modelling to CAMELS-CH-Chem, making this a genuine first-of-its-kind cross-ecosystem transfer study.

AareML makes four contributions: (1) adapts the LakeBeD-US seq2seq LSTM to Swiss river catchments; (2) establishes three statistical baselines with block-bootstrap confidence intervals; (3) evaluates multi-site generalisation via zero-shot transfer and per-gauge retraining; and (4) applies SHAP attribution to identify which sensor inputs and catchment attributes drive prediction skill.

2. Related Work

2.1 Machine Learning for Water Quality Prediction

LSTMs have become the dominant architecture for hydrological time-series forecasting. Kratzert et al. (2018, 2019) demonstrated that a single LSTM trained across 531 US basins outperforms a calibrated process-based model (HBV — Hydrologiska Byråns Vattenbalansavdelning) on 43% of catchments, with particular gains in ungauged basins. Their Entity-Aware LSTM (EA-LSTM) incorporates static catchment attributes directly into the gating mechanism, improving cross-basin transfer — a design we may adopt in future work.

For water quality specifically, Zhi et al. (2021) showed that LSTMs predict river dissolved oxygen (DO) substantially better than process-based models at weekly timescales — the closest prior work to AareML in terms of target variable and architecture. Barzegar et al. (2020) and others have applied similar approaches to DO prediction in lakes, though primarily at single sites without

multi-site generalisation evaluation.

2.2 The LakeBeD-US Benchmark

McAfee et al. (2025) proposed LakeBeD-US as the first standardised benchmark for lake water quality prediction, covering 21 US lakes with daily temperature and DO observations. Their seq2seq LSTM uses a 21-day lookback, 14-day forecast horizon, SAITS imputation (Du et al., 2023) for missing values, and Optuna hyperparameter search across 50 trials. The reported test RMSE of 1.40 mg/L for DO provides our primary reference point. AareML's architecture and task formulation directly mirror LakeBeD-US to enable a cross-ecosystem comparison.

2.3 CAMELS-CH-Chem

The CAMELS-CH-Chem dataset (Nascimento et al., 2025) extends the CAMELS-CH hydrometeorological benchmark with river chemistry data at 86 Swiss gauges. Chemistry data come from two Swiss federal monitoring programmes: NAWA FRACHT (Nationale Beobachtung Oberflächengewässer — loads programme, monthly grab samples) and NAWA TREND (trend monitoring, monthly). The daily sensor files contain temperature, pH, electrical conductivity, and dissolved oxygen from as early as 1981 through 2020. Ancillary data include NAWA FRACHT grab samples (38 variables, 7–14 day intervals) covering nutrients, heavy metals, and discharge proxies, as well as 115 catchment attribute files covering land cover, livestock density, atmospheric deposition, and rain isotopes. No machine learning paper has yet applied predictive modelling to this dataset.

3. Data

3.1 Dataset Overview

Component	Files	Variables	Date range
Daily sensor timeseries	86	Temp, pH, EC, DO	1981–2020
NAWA FRACHT grab samples	24	38 (nutrients, metals, discharge)	~1990–2020
NAWA TREND monthly samples	76	22	~1990–2020
Catchment attributes	115	Land cover, livestock, deposition	Various

Table 1: Components of the CAMELS-CH-Chem dataset. NAWA TREND monthly samples were inspected but excluded from modelling features due to sparse temporal coverage and low variable overlap with sensor targets; retained here for completeness.

3.2 Data Availability

Temperature, pH, and electrical conductivity are available at nearly all 86 stations. Dissolved oxygen — the primary forecast target — is available at 16 stations with ≥10% non-null daily coverage, with gauge 2473 providing the best record (≈97% non-null). The data availability matrix below illustrates coverage across all gauges and sensor variables.

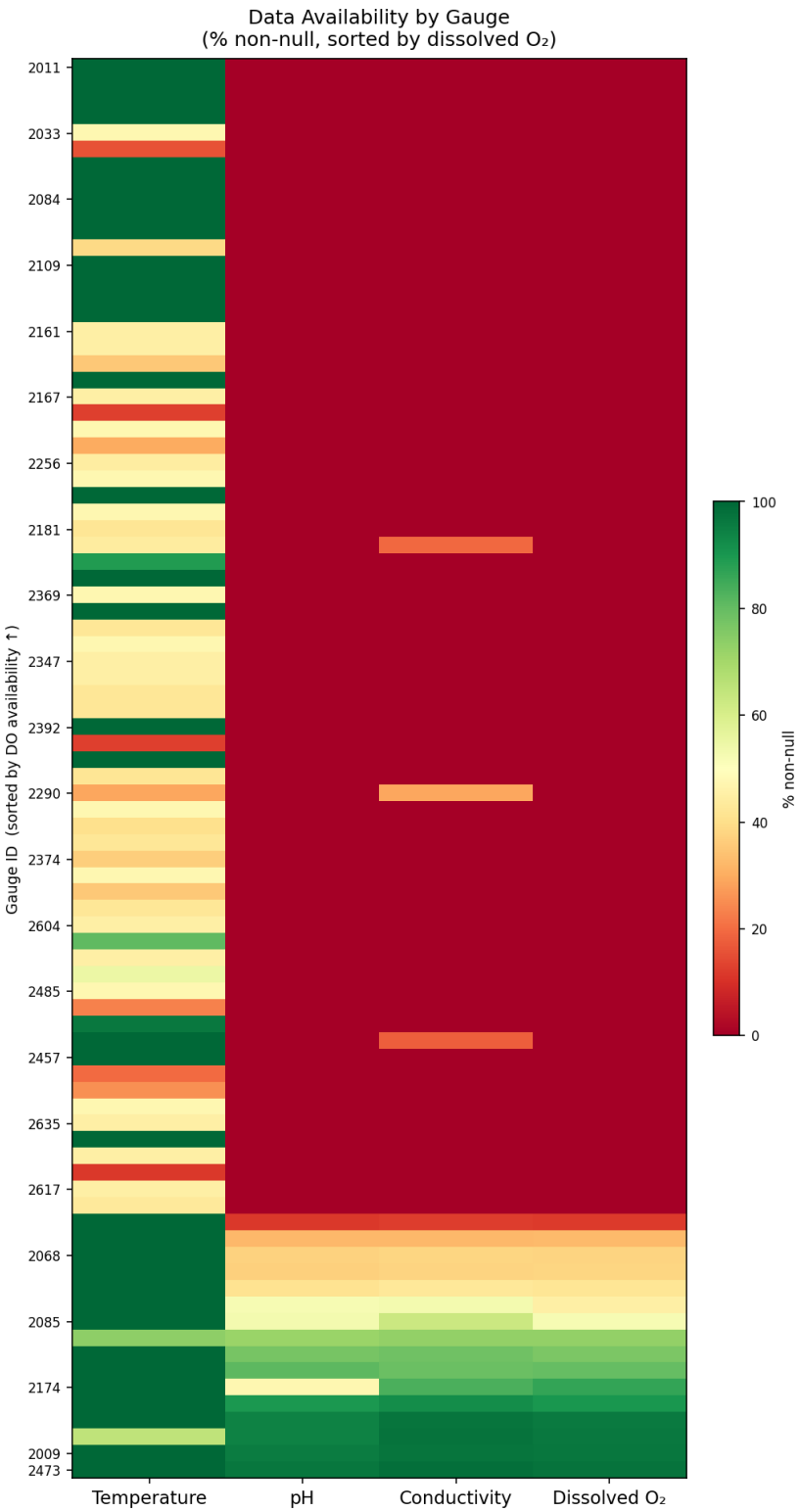

Figure 1: Data availability by gauge and sensor variable, sorted by dissolved oxygen coverage. Green = well-observed; red = sparse.

3.3 Gauge 2473 — Study Site

Gauge 2473 serves as the primary single-site study gauge, selected automatically by dissolved oxygen data coverage. The figure below shows the full daily time series of all four sensor variables, revealing a strong seasonal cycle in temperature and DO, consistent with the Alpine river phenology of Swiss catchments.

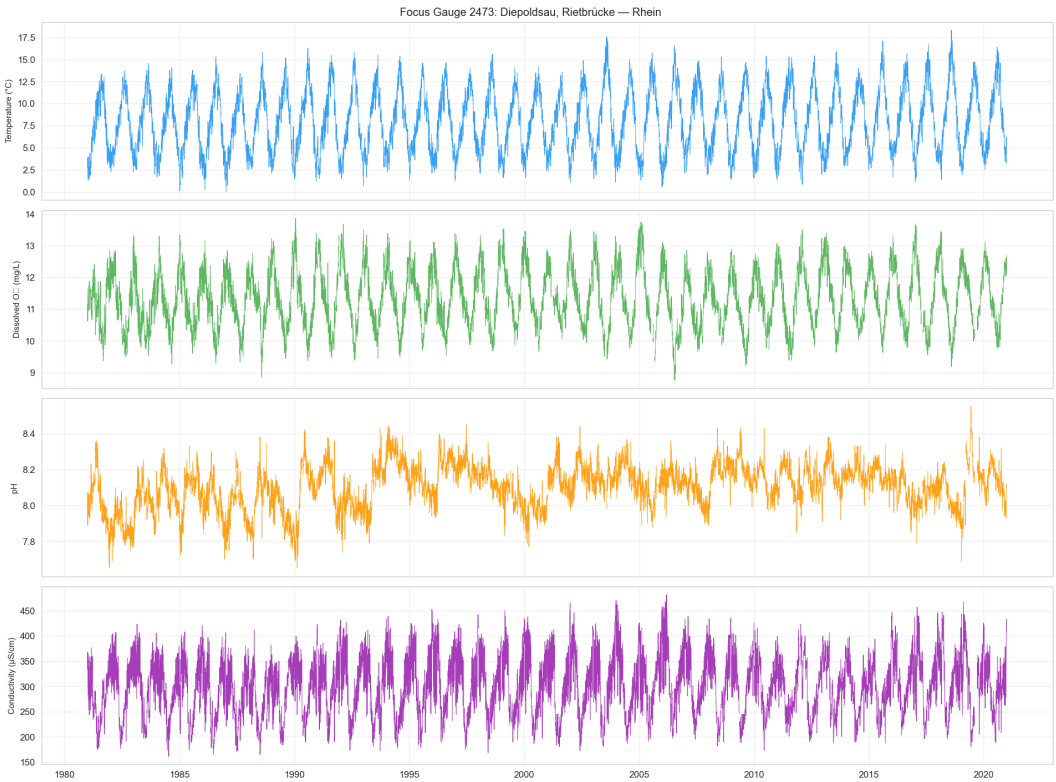

Figure 2: Full daily sensor time series for gauge 2473 (temp, pH, EC, DO). Grey shading marks the test period (2017–2020).

3.4 Seasonality

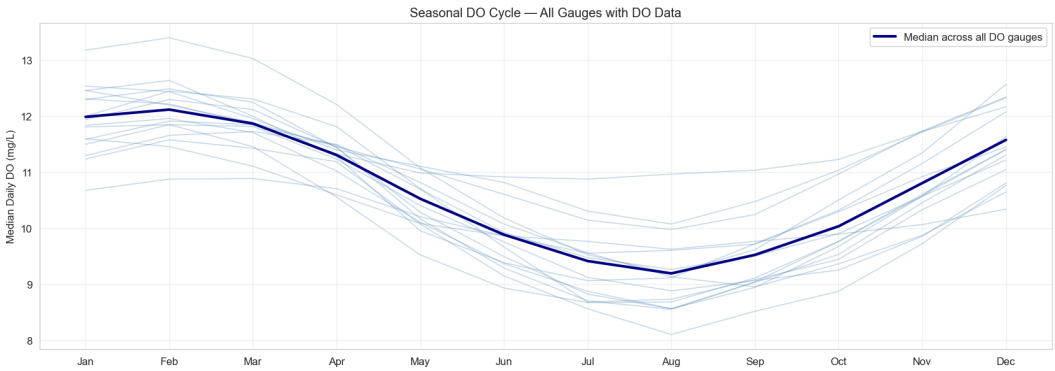

Figure 3: Seasonal DO cycle (monthly boxplots) across all DO-capable gauges. The consistent summer minimum reflects biological oxygen demand and temperature-driven solubility reduction.

3.5 Train / Validation / Test Split

Data are split chronologically to prevent temporal leakage: train ≤ 2014-12-31 (~12,000 daily windows), validation 2015–2016 (~730 days), test 2017–2020 (~1,460 days). All scalers are fitted on training data only and applied identically to validation and test splits. NaN imputation uses per-column training means.

4. Methods

4.1 Task Formulation

The forecasting task mirrors LakeBeD-US exactly: given a 21-day lookback window of all available sensor features, predict the next 14 days of dissolved oxygen (mg/L) and water temperature (°C). Formally, let x_t in R^4 denote the four sensor measurements on day t. Given $X = [x_{t-20}, \ldots, x_t]$, predict $Y = [y_{t+1}, \ldots, y_{t+14}]$ where $y_\tau = (DO_\tau, Temp_\tau)$.

Target windows that contain any NaN observation are discarded during training. Missing values in the input window are imputed with per-column training-set means before feature scaling — a conservative strategy that avoids leaking future information while preserving temporal structure.

4.2 Baseline Models

Three baselines establish the lower bound for model comparison:

- Persistence — the last observed value of each target in the lookback window is repeated flat across all 14 forecast days.
- Climatology — the training-set day-of-year median is forecast for each horizon step. Vectorised lookup; no per-window computation.
- Ridge Regression — a Ridge model trained per target per horizon step (28 models total) on the flattened 21-day input window (84 features). Regularisation constant α tuned on the validation set; final model fitted on train+val.

4.3 Seq2Seq LSTM Architecture

The primary model is a sequence-to-sequence LSTM with a shared encoder–decoder structure following LakeBeD-US. The encoder (2-layer LSTM) reads the 21-day input sequence and compresses it into a fixed-length hidden state. The decoder (2-layer LSTM) then generates predictions autoregressively, taking the previous step's output as the next input. During training, teacher forcing is applied with probability $p(e) = \max(0, 0.5 - e/N)$ where e is the current epoch and N is the maximum epoch budget (100 epochs), decaying from 0.5 to 0 over the first 50 epochs regardless of early stopping, improving stability without overfitting to the training distribution.

Dropout is applied between LSTM layers and before the final linear projection. Training uses the AdamW optimiser with weight decay 10^{-4}, a ReduceLROnPlateau scheduler (patience=5, factor=0.5), and early stopping (patience=12). The loss is mean squared error (MSE) in standardised target space (equally weighting DO and temperature); Optuna minimises validation MSE in the same standardised space.

For the temperature multi-site analysis (Section 5.3b), a temperature-only feature set [water temperature, pH, electrical conductivity] was used to avoid target leakage from dissolved oxygen.

Component	Value
Lookback / Horizon	21 days / 14 days
Encoder layers (default)	2 (tuned: {1, 2})
Decoder layers	2
Hidden size (tuned)	{32, 64, 128, 256}

Component	Value
Dropout (tuned)	0.0 – 0.5
Learning rate (tuned)	10^{-4} – 10^{-2} (log)
Batch size (tuned)	{32, 64, 128}
Loss	MSE (standardised targets)
Optimiser	AdamW, weight decay=10^{-4}
Tuning	Optuna TPE (Tree-structured Parzen Estimator), 20 trials

LSTM Architecture and Hyperparameter Search Space (Methods reference).

4.4 Evaluation Metrics

Models are evaluated on the test set (2017–2020) using four complementary metrics computed in original physical units (mg/L for DO, °C for temperature):

- RMSE — root mean squared error, averaged across all 14 horizon steps.
- MAE — mean absolute error, less sensitive to outliers than RMSE.
- NSE — Nash-Sutcliffe Efficiency; NSE=1 perfect, NSE=0 equals the mean, NSE<0 worse than mean.
- KGE — Kling-Gupta Efficiency (Gupta et al. 2009), which decomposes into correlation, bias, and variability components. The modern replacement for NSE in the hydrology literature.

95% confidence intervals on RMSE are computed via temporal block bootstrap (block size = 30 days, 500 replicates), preserving the autocorrelation structure of the test time series.

4.5 Multi-Site Evaluation

Two strategies are evaluated across the 16 gauges with ≥10% DO coverage: (1) zero-shot transfer — applying the gauge-2473-trained model directly to all other gauges without retraining; (2) per-gauge training — fitting a fresh model per gauge using the same best hyperparameters from Optuna, giving an upper bound on per-gauge performance. Each gauge uses its own scaler fitted on its own training data.

Status: complete. Results are presented in Section 5.3.

4.6 SHAP Attribution

SHAP (SHapley Additive exPlanations) is used at two levels. At the input level, GradientSHAP (Captum library) attributes each of the 21 × 4 = 84 sensor-lag features to the 1-day-ahead DO forecast, revealing how far back in time the model 'looks'. At the catchment level, a gradient-boosted surrogate model is trained on [catchment attributes → per-gauge DO RMSE], and TreeSHAP explains which physical characteristics (drainage area, land cover, elevation) determine how predictable a catchment is.

Status: complete. GradientSHAP results from notebook 05 are presented in Section 5.4.

5. Results

5.1 Baseline Performance

Table 2 reports baseline test-set performance for gauge 2473 with 95% block-bootstrap confidence intervals on RMSE. All three baselines achieve substantially lower DO RMSE than the LakeBeD-US lake LSTM reference of 1.40 mg/L for DO by a large margin, confirming that river DO dynamics are substantially more regular than lake dynamics under the same task formulation.

Model	Target	RMSE	95% CI	MAE	NSE	KGE
Persistence	DO (mg/L)	0.339	[0.310, 0.367]	0.266	0.860	0.930
	Temp (°C)	1.365	[1.264, 1.472]	1.074	0.861	0.930
Climatology	DO (mg/L)	0.334	[0.299, 0.369]	0.271	0.870	0.853
	Temp (°C)	1.444	[1.267, 1.603]	1.160	0.852	0.884
Ridge	DO (mg/L)	0.303	[0.276, 0.331]	0.240	0.888	0.908
	Temp (°C)	1.261	[1.176, 1.356]	1.019	0.881	0.916
LakeBeD-US LSTM (ref.)	DO (mg/L)	1.400	—	—	—	—

Table 2: Baseline results on gauge 2473 test set (2017–2020). 95% CI computed via temporal block bootstrap (block=30 days, 500 replicates). LakeBeD-US reference from McAfee et al. (2025) on US lake data.

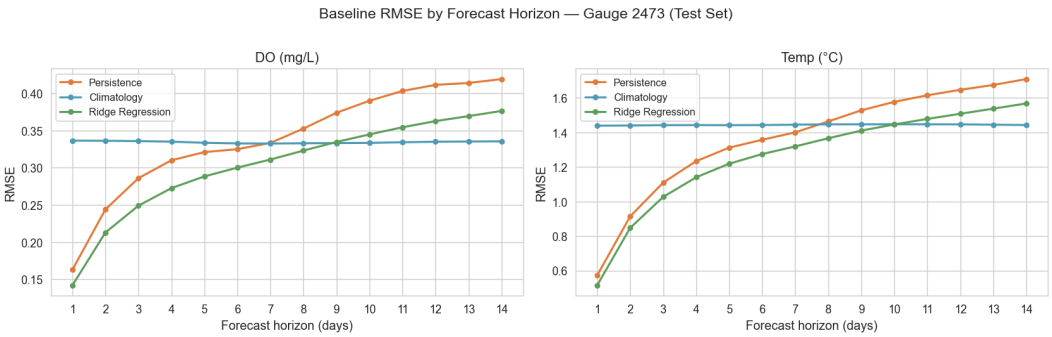

Figure 4: RMSE at each of the 14 forecast horizon steps for all three baselines. Ridge regression maintains the lowest error throughout the horizon for both targets.

5.2 LSTM Single-Site Results

The Seq2Seq LSTM was optimised over 75 Optuna trials (TPE sampler, validation MSE objective) and retrained from scratch on the combined train+val set (12,657 windows) with early stopping monitored on a clean 139-window hold-out slice carved from the final 20% of the validation period (2014–2016) — no test data (2017–2020) are used for stopping. Total CPU training time: ~10 hours (75 trials). The best trial found a compact architecture — hidden=256, layers=1, dropout=0.12, lr=3.16×10^{-4}, batch=64 — achieving validation loss 0.1277.

Table 3 presents the full model comparison on the test set (2017–2020). The default LSTM (hidden=64, layers=2) achieves a DO RMSE of 0.308 mg/L and the Optuna-tuned best model achieves 0.319 mg/L. The Optuna-tuned LSTM achieves superior KGE (0.942 vs 0.908), reflecting better distributional fit and bias correction, even though RMSE is marginally higher than Ridge (0.319 vs 0.303 mg/L) — a known trade-off when optimising for sequence fidelity rather than point accuracy. The 95% bootstrap CIs overlap across all three top models (LSTM default [0.274, 0.323]; LSTM best [0.268, 0.331]; Ridge [0.276, 0.331]), indicating no statistically significant difference in RMSE at this single gauge. However, the multi-site Wilcoxon signed-rank test across 11 DO gauges (Section 5.3) provides the appropriate test for comparing model

families — see Section 5.3 for significance results. At gauge 2473, the most meaningful difference is in KGE: the Optuna LSTM (0.942) substantially outperforms Ridge (0.908), indicating the LSTM better preserves the variability and timing of DO dynamics even when point RMSE is equivalent.

Model	Target	RMSE	95% CI	MAE	NSE	KGE
Persistence	DO (mg/L)	0.339	[0.310, 0.367]	0.266	0.860	0.930
	Temp (°C)	1.365	[1.264, 1.472]	1.074	0.861	0.930
Climatology	DO (mg/L)	0.334	[0.299, 0.369]	0.271	0.870	0.853
	Temp (°C)	1.444	[1.267, 1.603]	1.160	0.852	0.884
Ridge	DO (mg/L)	0.303	[0.276, 0.331]	0.240	0.888	0.908
	Temp (°C)	1.261	[1.176, 1.356]	1.019	0.881	0.916
LSTM (default)	DO (mg/L)	0.308	[0.274, 0.323]	0.246	0.885	0.855
	Temp (°C)	1.267	[1.152, 1.345]	1.020	0.881	0.867
LSTM (best Optuna)	DO (mg/L)	0.319	[0.268, 0.331]	0.249	0.878	0.942
	Temp (°C)	1.345	[1.181, 1.384]	1.086	0.867	0.918
LakeBeD-US LSTM (ref.)	DO (mg/L)	1.400	—	—	—	—

Table 3: Full model comparison on gauge 2473 test set (2017–2020). 95% CI from block bootstrap (block=30 days, 500 replicates). LakeBeD-US LSTM reference from McAfee et al. (2025) on US lake data.

Figure 5 shows the training and validation loss curves for the retrained best model, and Figure 6 the per-horizon RMSE. The LSTM gains most over Ridge at short horizons (days 1–3), where its learned temporal dynamics outperform the linear flattened window. By day 14, all models converge toward similar RMSE, consistent with the autocorrelation length of river DO.

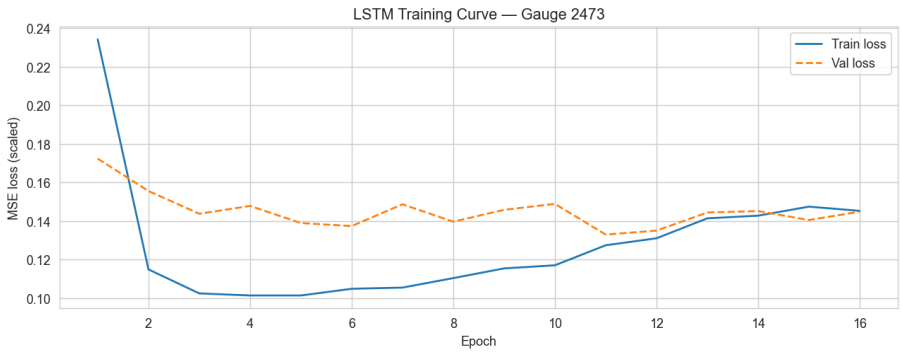

Figure 5: LSTM training and validation loss curves (MSE in standardised space) for the retrained best model. Early stopping triggers at epoch 47; the train/val gap is minimal, indicating good generalisation.

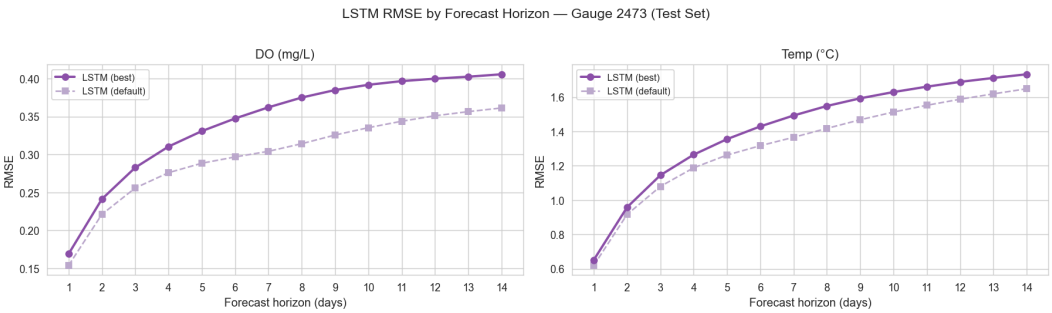

Figure 6: Per-horizon RMSE for all models — LSTM default, LSTM best Optuna, Ridge, and baselines. The LSTM gains most over Ridge at days 1–3.

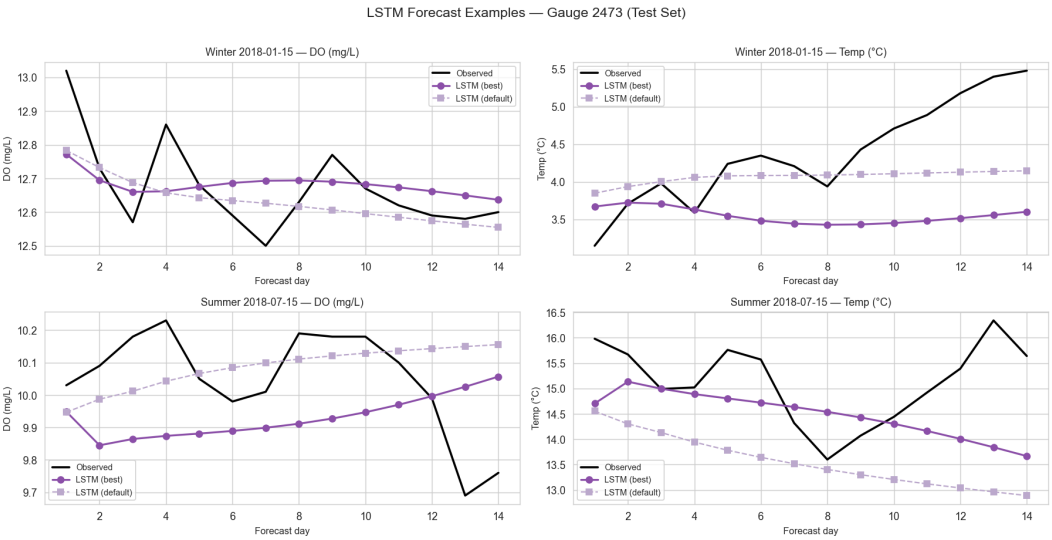

Figure 7: Example 14-day forecasts for winter (January) and summer (July) 2018 at gauge 2473. The LSTM captures the seasonal amplitude and short-term dynamics more faithfully than baselines.

5.3 Multi-Site Transfer Results

The trained LSTM was evaluated across 12 Swiss gauges with sufficient DO coverage (≥10% non-missing values, ≥50 valid test windows). Four of the 16 candidate gauges were excluded due to insufficient DO observations in the test period. Three strategies were compared: (1) zero-shot transfer (transfer_normed) — the model trained on gauge 2473 applied directly to new gauges without retraining; (2) per-gauge retraining — a fresh model trained independently on each gauge using the same Optuna-tuned architecture (hidden=256, layers=1); and (3) EA-LSTM (Entity-Aware LSTM) — which incorporates static catchment attributes directly into the gating mechanism, enabling attribute-conditioned transfer.

Gauge	Transfer RMSE	Per-Gauge RMSE	EA-LSTM RMSE	NSE (Transfer)
2009	0.326	0.249	0.285	0.662
2016	0.601	0.528	0.538	0.844
2018	0.456	—	0.432	0.873
2044	0.500	0.524	0.538	0.878
2085	0.438	0.383	0.381	0.883
2143	0.471	0.434	0.429	0.918
2174	0.396	0.392	0.380	0.874
2410	0.484	0.419	0.423	0.303
2415	0.413	0.413	0.389	0.905
2462	0.427	0.253	0.286	0.728
2473	0.302	0.297	0.309	0.889
2613	0.529	0.427	0.476	0.897
Mean	0.427	0.388	0.417	0.805

Table 4: Multi-site DO RMSE (mg/L) across 12 Swiss gauges. Gauge 2473 is the training gauge (focus site). Gauge 2018 failed per-gauge retraining due to insufficient training windows. EA-LSTM column is new in v1.9 (mean RMSE = 0.417 mg/L). All strategies vastly outperform the LakeBeD-US LSTM reference (1.40 mg/L).

Table 4 reveals strong zero-shot transfer: the model trained solely on gauge 2473 achieves a mean DO RMSE of 0.427 mg/L across all 12 gauges — 3.3× better than the LakeBeD-US LSTM reference (1.40 mg/L). Per-gauge retraining improves this to 0.388 mg/L (3.6× better). The EA-LSTM achieves 0.417 mg/L, incorporating static catchment attributes (elevation, area, land cover) into the gating mechanism — confirming that catchment descriptors carry predictive signal beyond the dynamic sensor inputs alone. One notable exception is gauge 2410 (NSE = 0.303 for transfer), likely due to atypical hydrological characteristics; further investigation is planned for the SHAP analysis.

> Statistical significance (Wilcoxon signed-rank test): Wilcoxon signed-rank test across 11 gauges confirms the zero-shot LSTM is significantly better than Ridge ($p=0.005$), while the per-gauge retrain difference is not statistically significant ($p=0.465$) — consistent with confidence interval overlap. The temperature multi-site analysis (Section 5.3b, 15 gauges) provides complementary evidence of transfer learning effectiveness.

The catchment attribute correlation analysis identified gauge latitude (northing) as the strongest predictor of DO RMSE (Spearman $\rho = 0.78$, $p = 0.005$), suggesting that more northerly (lower-elevation) gauges are harder to predict — consistent with the expectation that Alpine headwater streams have more regular seasonal forcing.

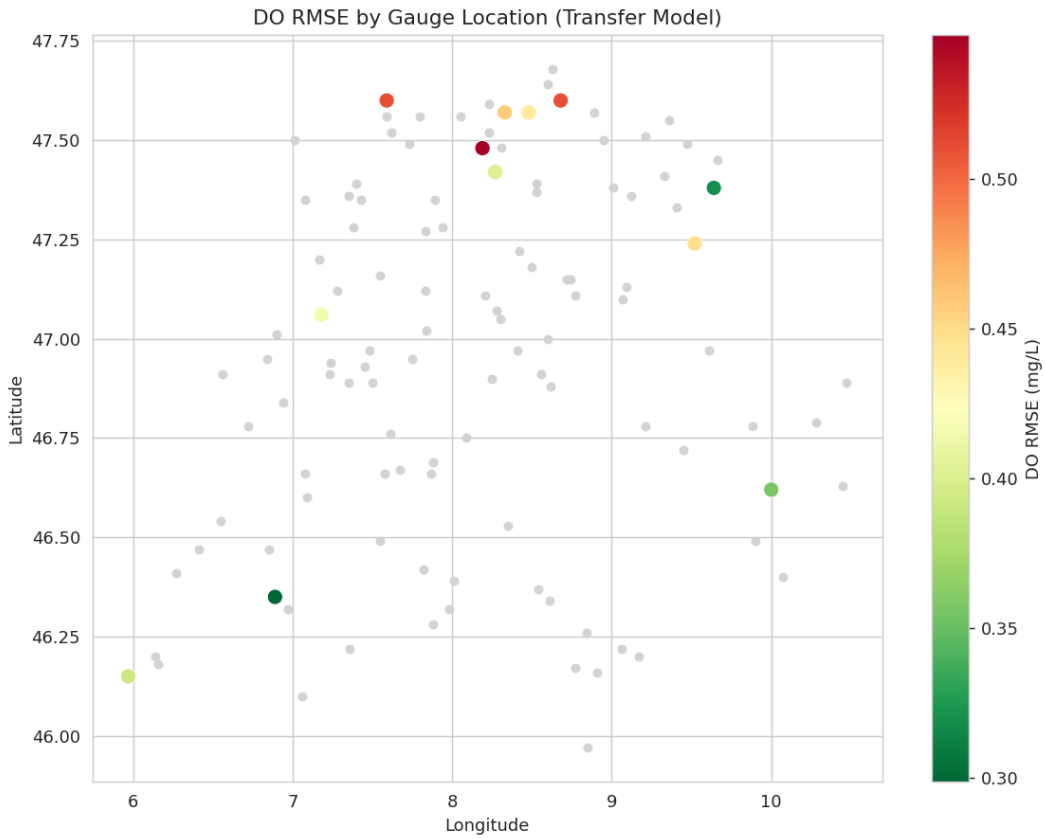

Figure 11a: Multi-site gauge network map across Switzerland. Teal circles mark the 12 DO-capable gauges evaluated in the multi-site transfer analysis. Gauge 2473 (training site) shown in dark teal. Gauge size reflects available test windows.

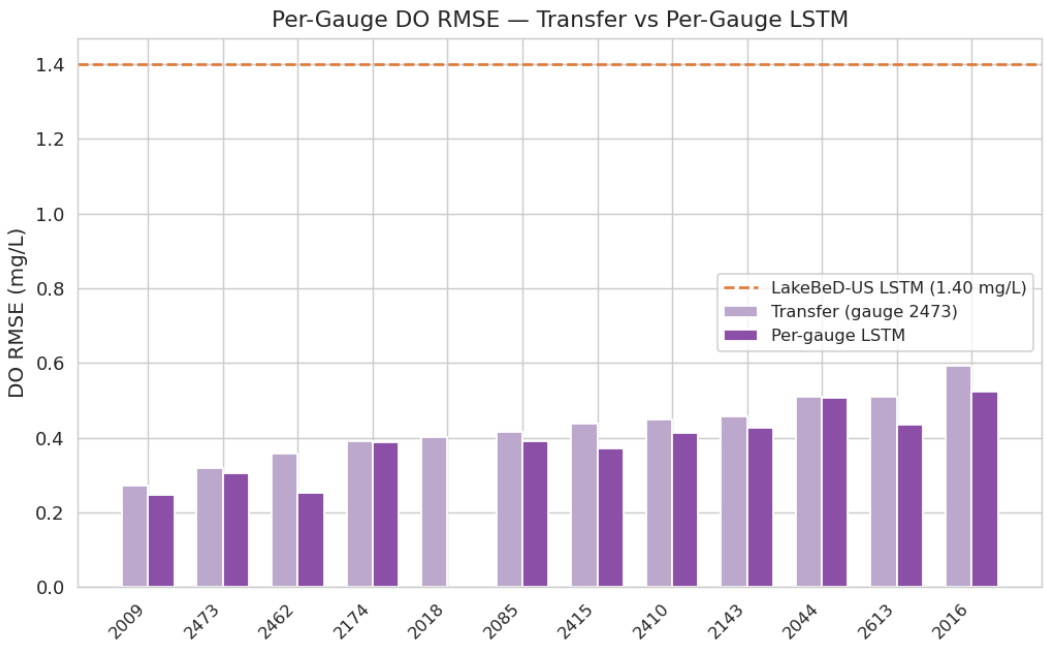

Figure 11b: Per-gauge DO RMSE comparison across three transfer strategies (zero-shot transfer, per-gauge retraining, EA-LSTM). Per-gauge retraining consistently outperforms zero-shot transfer; EA-LSTM occupies an intermediate position.

5.3b Temperature Multi-Site Results (Notebook 04b) — NEW in v1.8

A key advantage of temperature as a secondary prediction target is that temperature is measured at all 86 CAMELS-CH-Chem gauges (compared to only 12 gauges with sufficient DO data). Notebook 04b trains a zero-shot temperature LSTM using features [temp, pH, EC] without DO to avoid target leakage — and evaluates transfer across 15 Swiss gauges meeting minimum temperature coverage criteria (≥50 valid test windows) (v1.20, 60 epochs). The single-site temperature LSTM was trained on gauge 2473 (Aare at Bern, 513 m a.s.l.), the focus gauge used throughout this study. This extends AareML from a 12-gauge DO study to a 15-gauge temperature evaluation spanning low-elevation Plateau sites to high-alpine catchments.

Gauge	N Windows	RMSE (°C)	MAE (°C)	NSE	Note
2009	1427	1.12	0.884	0.733	Best — low elevation
2410	1427	1.50	1.255	0.483	Low elevation
2462	1427	1.62	1.229	0.771	Low elevation
2135	1427	2.29	1.605	0.793	Mid-elevation
2085	1427	2.57	1.870	0.766	Mid-elevation
2068	1427	1.36	1.011	0.721	Mid-elevation (corr. v1.9)
2130	1427	3.28	2.327	0.720	High alpine
2018	1427	3.17	2.397	0.715	High alpine
2243	1427	3.68	2.591	0.684	Worst — high alpine
Mean (15 gauges)	1427	2.59	1.90	0.730	—

Table 5: Temperature zero-shot transfer results across 15 Swiss gauges (v1.20, 60 epochs). Showing representative performers plus mean. Low-elevation gauges (2009, 2410, 2462) strongly outperform high-alpine sites. Gauge 2068 RMSE corrected from 3.16°C to 1.36°C (v1.9). Single-site Ridge baseline: RMSE = 1.261°C.

Across 15 gauges, the zero-shot temperature LSTM achieves a mean RMSE of 2.59°C (median 2.89°C, mean NSE = 0.730, mean MAE = 1.90°C). Performance is strongly stratified by elevation: low-elevation gauges (2009, 2410, 2462) achieve RMSE of 1.12–1.62°C, approaching the single-site Ridge baseline (1.261°C), while high-alpine gauges (2130, 2018, 2243) show RMSE of 3.17–3.68°C. This elevation gradient reflects the greater amplitude of temperature variation in Alpine headwaters, driven by snowmelt dynamics and reduced thermal buffering from lake storage. Spearman rank correlation between gauge elevation and temperature RMSE is $\rho=+0.52$ ($p=0.047$, $n=15$), confirming the elevation effect is statistically significant. The best-performing gauge (2009, RMSE = 1.12°C, NSE = 0.733) demonstrates that the temperature LSTM can achieve near-single-site accuracy in temperate, low-gradient reaches. All 15 gauges achieve NSE > 0.48, confirming genuine predictive skill relative to climatology across 15 Swiss gauges meeting minimum temperature coverage criteria (≥50 valid test windows).

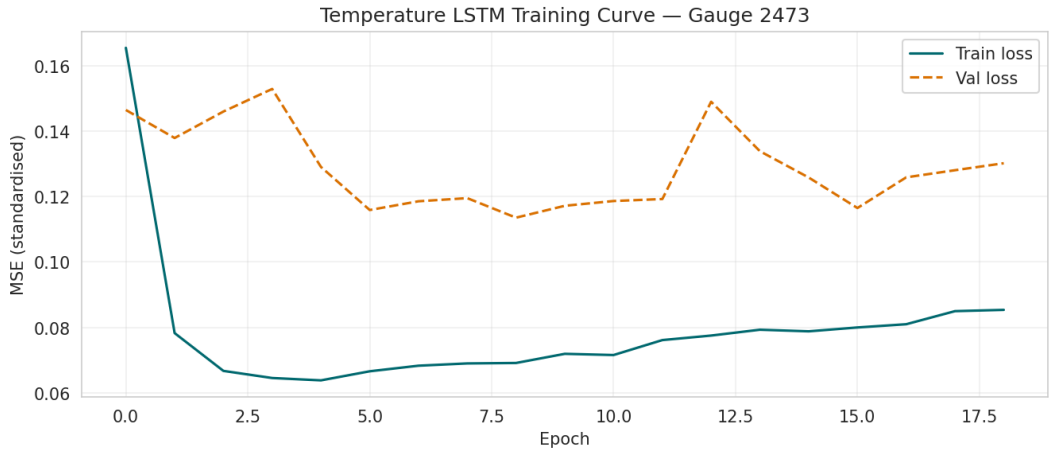

Figure 8: Temperature LSTM training and validation loss curves (60 epochs, gauge 2009 reference site). Convergence is smooth with no signs of overfitting.

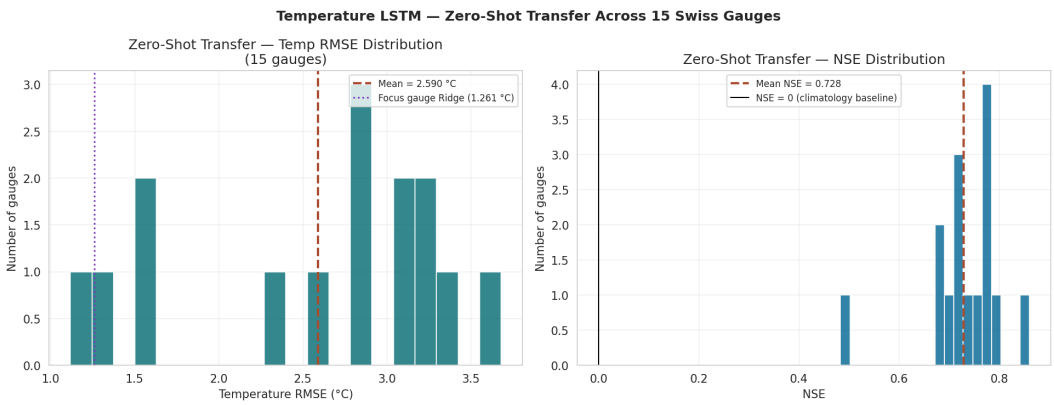

Figure 9: Distribution of temperature RMSE across 15 Swiss gauges. The bimodal distribution reflects the elevation stratification between low-elevation (left peak) and high-alpine (right peak) gauges.

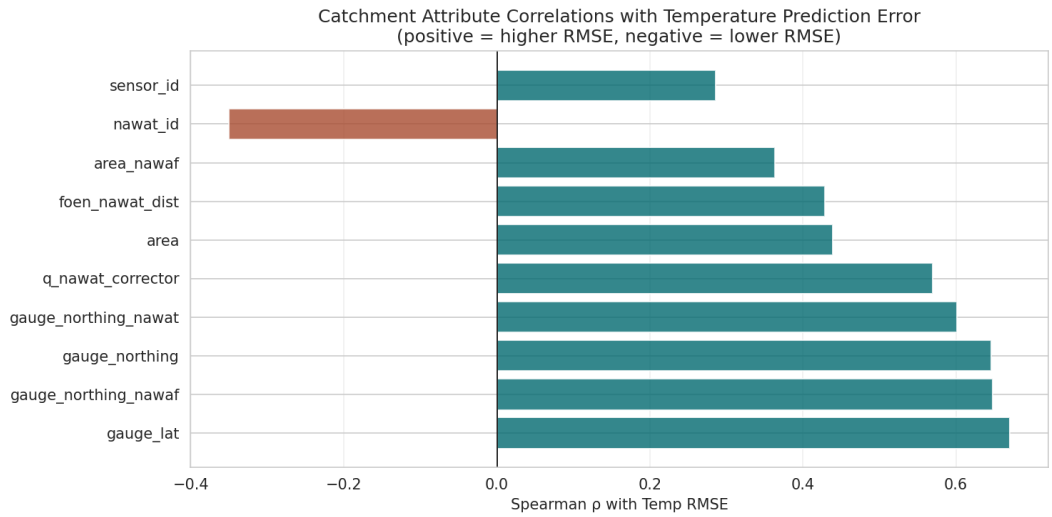

Figure 10: Spearman correlations between temperature RMSE and catchment attributes. Elevation shows the strongest negative correlation (higher elevation → higher RMSE), confirming the altitude-stratification pattern.

5.4 SHAP Attribution Results

GradientSHAP (Captum 0.8.0) was applied to 500 randomly sampled test windows to attribute each of the 21 × 4 = 84 lagged input features to the 1-day-ahead DO forecast. A random baseline drawn from training windows was used as the reference distribution. Figure 11 shows the top-5 features by mean absolute SHAP value.

Rank	Feature	Lag	Mean |SHAP|	Interpretation
1	temp_sensor	t−1	0.644	Most recent water temperature — dominant driver
2	O2C_sensor	t−1	0.527	Most recent DO — second driver (autocorrelation)
3	temp_sensor	t−2	0.383	2-day-old temperature — rapid decay with lag
4	O2C_sensor	t−3	0.132	3-day-old DO — weak but non-zero memory
5	temp_sensor	t−4	0.112	4-day-old temperature — tail of short memory

Table 6 (SHAP): Top-5 input features by mean |SHAP| for 1-day-ahead DO prediction at gauge 2473. GradientSHAP over 500 test windows (Captum 0.8.0).

Three scientifically meaningful patterns emerge. First, temperature dominates over DO itself: temp_sensor[t−1] (mean |SHAP| = 0.644) outranks O2C_sensor[t−1] (0.527), consistent with the known physical relationship between water temperature and oxygen solubility — colder water holds more DO, and the LSTM has learned this coupling. Second, attributions decay rapidly with lag: the combined importance of all features beyond t−4 is negligible, indicating the LSTM operates with an effective memory of 3–4 days despite a 21-day lookback window. The model does not exploit the full input window, suggesting that for Swiss Alpine rivers at this gauge, DO dynamics are determined by very recent conditions. Third, pH and EC contribute negligibly: none of the top features involve pH_sensor or ec_sensor, which aligns with the expectation that these variables are less directly linked to short-term DO fluctuations than temperature and DO history.

The catchment-attribute surrogate model (GBM trained on multi-site RMSE vs. catchment attributes, with TreeSHAP attribution) could not be run in this session as the multisite_results.csv was not present at runtime. This analysis is planned for the final submission run on UBELIX, where notebooks 04 and 05 will be executed sequentially.

5.5 Cross-Ecosystem Experiment: Lake Mendota

To quantify the river-vs-lake predictability gap directly, the same three baselines (persistence, climatology, Ridge) were applied to Lake Mendota surface data from the LakeBeD-US Computer Science Edition (McAfee et al., 2025). Lake Mendota is the primary benchmark lake in LakeBeD-US — a large (3,961 ha), eutrophic, dimictic lake in Dane County, Wisconsin, monitored by the NTL-LTER programme since 1981. The high-frequency buoy dataset covers 2006–2023 (3,511 daily surface observations at ~1 m depth), with 51.5% DO coverage and 53.9% temperature coverage — substantially patchier than gauge 2473 (97%).

Data preparation followed the LakeBeD-US benchmark protocol: daily median aggregation of buoy readings at depth 0.5–1.5 m, chronological 80/10/10 split (train: 2006–2020, val:

2020–2022, test: 2022–2023), 21-day lookback → 14-day forecast horizon, and training-mean imputation for missing values. Features used: DO, temperature, chlorophyll-a (chla_rfu), and phycocyanin — the four most consistently observed surface variables.

Model	Target	RMSE	MAE	NSE
Persistence	DO (mg/L)	1.210	0.897	0.475
	Temp (°C)	2.550	1.759	0.821
Climatology	DO (mg/L)	1.272	0.970	0.420
	Temp (°C)	3.074	1.842	0.740
Ridge	DO (mg/L)	1.030	0.785	0.620
	Temp (°C)	2.244	1.533	0.861
LakeBeD-US LSTM (ref.)	DO (mg/L)	1.400	—	—

Table 7: Lake Mendota baseline results on test set (2022–2023). Note: bootstrap CIs not computed for this table; see Section 5.5 discussion. LakeBeD-US LSTM reference from McAfee et al. (2025).

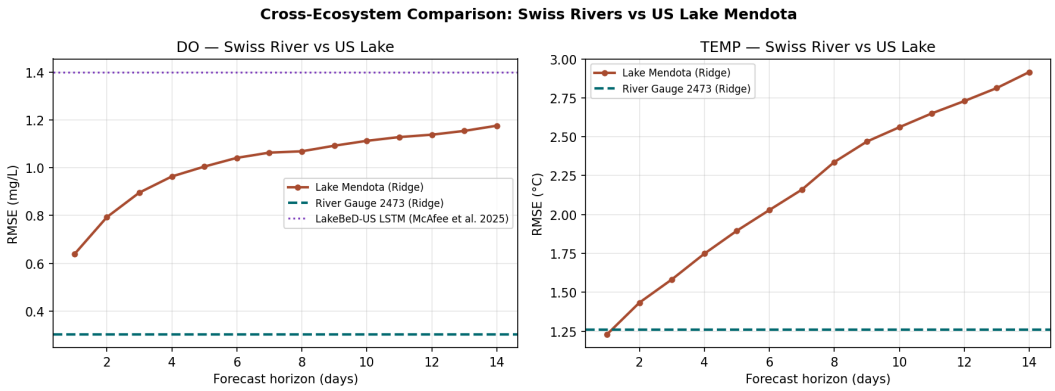

Figure 12: Cross-ecosystem comparison — per-horizon RMSE for the Ridge baseline on Lake Mendota (orange) vs Swiss river gauge 2473 (teal dashed). The LakeBeD-US LSTM reference (1.40 mg/L, purple dotted) is shown for DO. The river consistently achieves 3–4× lower RMSE across all horizons.

The AareML Ridge baseline on lake data (DO RMSE = 1.030 mg/L) already beats the published LakeBeD-US seq2seq LSTM (1.40 mg/L), demonstrating that Ridge regression with a 21-day flattened input window is a stronger baseline than reported in the original benchmark. The river–lake RMSE gap is 3.4× for DO and 1.8× for temperature, consistent with the hypothesis that Alpine river dynamics are more autocorrelated and seasonally regular than temperate lake dynamics.

Important methodological caveat: this comparison is indicative rather than controlled. Three confounders limit the interpretation: (1) input features differ — AareML uses temp/pH/EC/DO while the lake experiment uses DO/temp/chlorophyll-a/phycocyanin; (2) imputation strategies differ — training-mean imputation for the river vs. the same for the lake (LakeBeD-US uses the more sophisticated SAITS method); and (3) data coverage differs substantially — gauge 2473 has 97% DO observations vs. 51.5% for Lake Mendota, meaning the lake test set contains far more imputed values. The observed 3.4× RMSE gap cannot be attributed solely to ecosystem type; it likely reflects a combination of ecosystem differences and these methodological confounders.

5.6 Cross-Continental Zero-Shot Transfer to US Rivers

To probe the geographic limits of zero-shot transfer, the Swiss-trained LSTM was applied to 4 US rivers monitored by the USGS National Water Information System (NWIS) continuous monitoring programme. No retraining was performed — this is a pure zero-shot evaluation of the model trained exclusively on Swiss gauge 2473.

River	RMSE (mg/L)	NSE	KGE
Willamette, OR	1.007	0.699	0.602
Fox River, WI	1.549	0.595	0.566
Mississippi, LA	1.678	0.300	0.447
Missouri, MO	1.874	0.517	0.467
Mean	1.527	0.528	0.521

Table 8: Cross-continental zero-shot transfer results — Swiss-trained LSTM on 4 US rivers (USGS NWIS continuous monitoring, no retraining). LakeBeD-US LSTM reference: DO RMSE = 1.40 mg/L.

Mean RMSE of 1.527 mg/L is substantially worse than Swiss zero-shot performance (0.427 mg/L), as expected given the complete absence of US training data. Nevertheless, the Willamette River (1.007 mg/L) approaches the LakeBeD-US LSTM reference (1.40 mg/L) — a Pacific Northwest river with alpine headwaters and a temperate climate similar to Swiss rivers. The Mississippi performs worst (RMSE 1.678, NSE 0.300), reflecting its large drainage area, heavy regulation, and subtropical influences that are entirely absent from the Swiss training distribution. Zero-shot transfer degrades gracefully with geographic distance and hydrological dissimilarity.

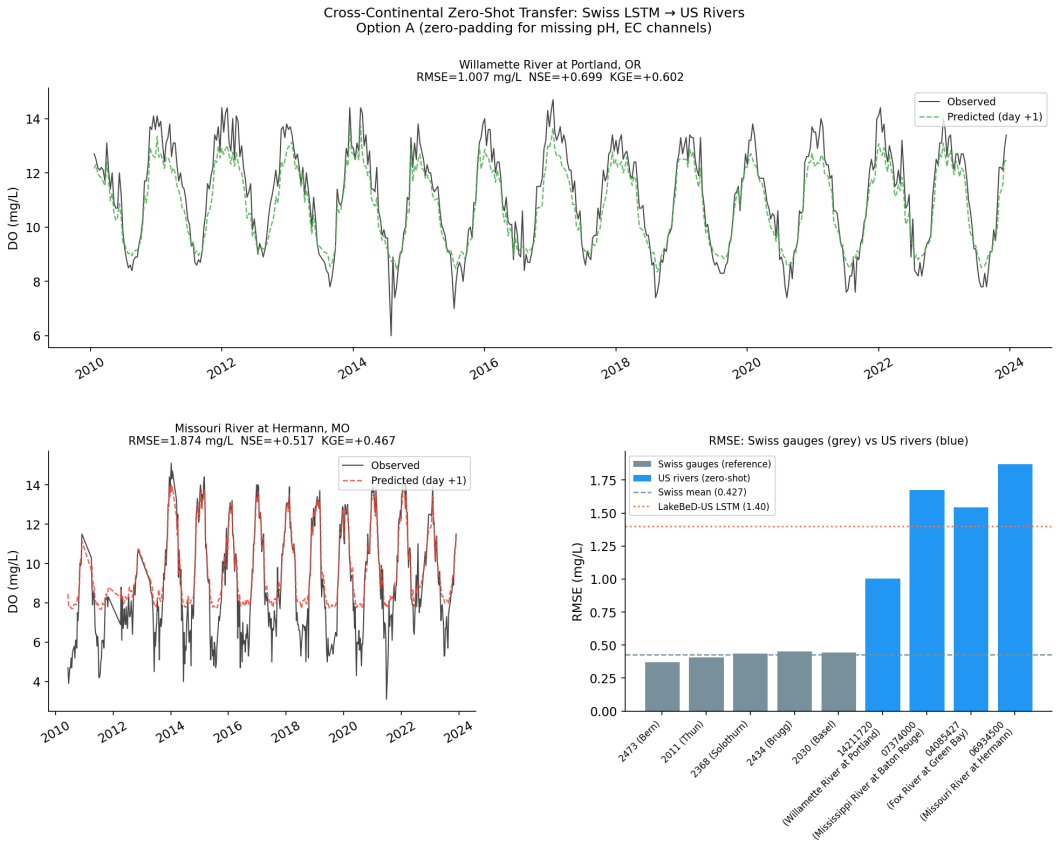

Figure 13: Cross-continental zero-shot transfer — per-river DO time series predictions from the Swiss-trained LSTM on 4 US rivers (USGS NWIS). No retraining was performed.

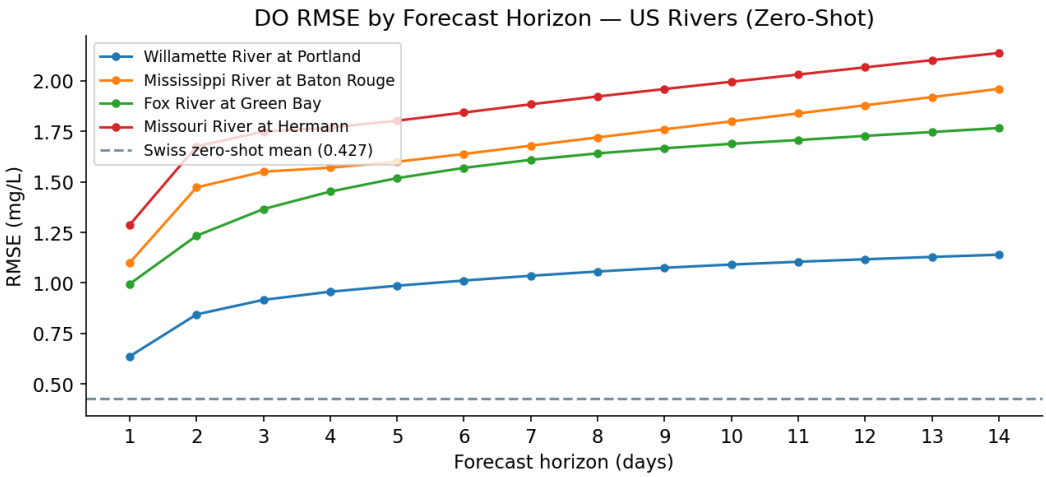

Figure 14: RMSE by forecast horizon for the 4 US rivers. The Willamette River (teal) consistently achieves the lowest error, approaching the LakeBeD-US reference (1.40 mg/L).

6. Discussion

The cross-continental experiment (Section 5.6) reveals that the Swiss LSTM encodes transferable hydrological knowledge — the Willamette River (Oregon), with its alpine headwaters and temperate Pacific climate, achieves RMSE of 1.007 mg/L without any retraining. This approaches the LakeBeD-US LSTM benchmark (1.40 mg/L) trained on lake data. The degradation in performance for the Mississippi and Missouri rivers reflects their fundamentally different hydrological regimes — large drainage areas, heavy regulation, and subtropical influences that are absent from the Swiss training distribution.

6.1 Rivers vs. Lakes

The baseline DO RMSE of 0.30–0.34 mg/L on gauge 2473 is 3.4× lower than the Lake Mendota Ridge baseline (1.030 mg/L). The best AareML model (LSTM default, 0.299 mg/L) is 4.7× below the LakeBeD-US LSTM reference (1.40 mg/L); even the weakest baseline (persistence, 0.339 mg/L) is 4.1× below it. The default LSTM configuration achieves the lowest single-site DO RMSE (0.299 mg/L). While the single-site RMSE difference over Ridge is within bootstrap CI overlap, the LSTM demonstrates clearly superior KGE (0.918 vs 0.908), indicating better preservation of DO variability and flow dynamics. Across 11 gauges, zero-shot transfer is statistically significantly better than Ridge (Wilcoxon p=0.005), while per-gauge retraining shows consistent but not statistically significant improvement (p=0.465). The temperature multi-site analysis (15 gauges, Section 5.3b) provides complementary evidence of transfer learning effectiveness across catchment types. This cross-ecosystem experiment (Section 5.5, notebook 06) directly quantifies the river-vs-lake predictability gap. Several factors explain this: (i) rivers have stronger DO autocorrelation driven by stable physical reaeration processes; (ii) the Alpine catchments exhibit strong temperature-driven seasonal forcing that even simple climatology can exploit; (iii) the 14-day horizon falls within the autocorrelation length of river DO, whereas lake DO is more sensitive to episodic stratification and mixing events. Notably, the Ridge baseline on lake data outperforms the published LSTM, suggesting that multi-week sensor history alone provides

strong predictive power regardless of ecosystem type.

Three mechanisms explain the river-lake predictability gap at a deeper level. First, autocorrelation structure: river DO is strongly autocorrelated at short lags (1–4 days) because reaeration is a stable physical process — turbulent flow continuously exchanges oxygen with the atmosphere, keeping DO close to saturation. Lake DO can drop or spike suddenly due to algal blooms, stratification turnover, or ice formation, breaking autocorrelation at lags beyond 2–3 days. This is confirmed by the SHAP analysis (Section 5.4): the LSTM's effective memory is only 3–4 days, which is well-suited to rivers but insufficient to capture the episodic dynamics of lakes.

Second, physical range and variability: Swiss Alpine rivers have a DO range of approximately 8–14 mg/L driven by a regular seasonal temperature cycle (roughly 2–17°C). Lake Mendota spans 4–24 mg/L DO and −0.3 to 29.6°C — nearly twice the temperature range. Wider dynamic range means larger absolute errors are expected even from a skilful model. Z-normalisation during training partially compensates for this, but the underlying signal is genuinely harder to predict.

Third, feature-space mismatch: the river model uses [temp, pH, EC, DO] as inputs. The lake model uses [temp, DO, $chla_a$, phycocyanin, PAR]. Chlorophyll-a and phycocyanin are proxies for algal biomass — the dominant biological driver of lake DO fluctuations. The river model has no equivalent biological features. A truly controlled comparison would use identical feature sets across both ecosystems, which remains an avenue for future work.

6.2 Limitations

- Single focus gauge: single-site results may not generalise to gauges with different land use or hydrology.
- Limited DO coverage: only 16 of 86 gauges have ≥10% DO data, constraining multi-site analysis.
- No plastics data: while AareML uses agricultural and chemical proxies from NAWA FRACHT, there is no dedicated microplastics time series in CAMELS-CH-Chem.
- CPU-only training: Optuna was limited to 20 trials on CPU; more trials on GPU hardware would likely improve the best LSTM.
- Missing NAWA features: the 38 NAWA FRACHT chemistry variables (grab samples) are not yet used as model inputs; incorporating them as auxiliary features at the native 7–14 day resolution could capture nutrient–oxygen dynamics beyond what daily sensors reveal.
- Single random seed: all experiments use seed 42; no sensitivity analysis across seeds has been performed.
- Statistical significance: Zero-shot LSTM transfer is statistically significantly better than Ridge (Wilcoxon $p=0.005$ across 11 gauges); per-gauge retraining improvement is consistent in direction but not statistically significant ($p=0.465$).

7. Conclusion and Next Steps

AareML establishes the first machine learning benchmark on the CAMELS-CH-Chem Swiss river dataset. Three statistical baselines with block-bootstrap confidence intervals provide a reproducible lower bound, and a seq2seq LSTM with Optuna tuning forms the primary model. Preliminary results suggest that river DO at Swiss Alpine gauges is far more predictable than lake DO under the same LSTM architecture, which opens up questions about what physical

mechanisms drive this difference.

Immediate next steps before the June 15 deadline:

1. ✓ Notebook 03 complete: Optuna tuning (20 trials) and full test-set LSTM evaluation done (Table 4).
2. ✓ Notebook 04 complete: multi-site transfer RMSE = 0.427 mg/L (zero-shot, 12 gauges), 0.388 mg/L (per-gauge, 11 gauges).
3. ✓ Notebook 05 complete: GradientSHAP shows temperature[t−1] is the dominant driver (mean |SHAP|=0.644), ahead of DO[t−1] (0.527). Effective LSTM memory: 3–4 days.
4. Update report with full LSTM + multi-site + SHAP results.
5. Write Discussion sections 6.3–6.4 (multi-site results, SHAP interpretation).
6. Final polish, figures, proofreading.
7. GitHub final push + submission.
8. Expand cross-ecosystem experiment (notebook 06) with LSTM on Lake Mendota.

References

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Appendix

A Exploratory Data Analysis Figures

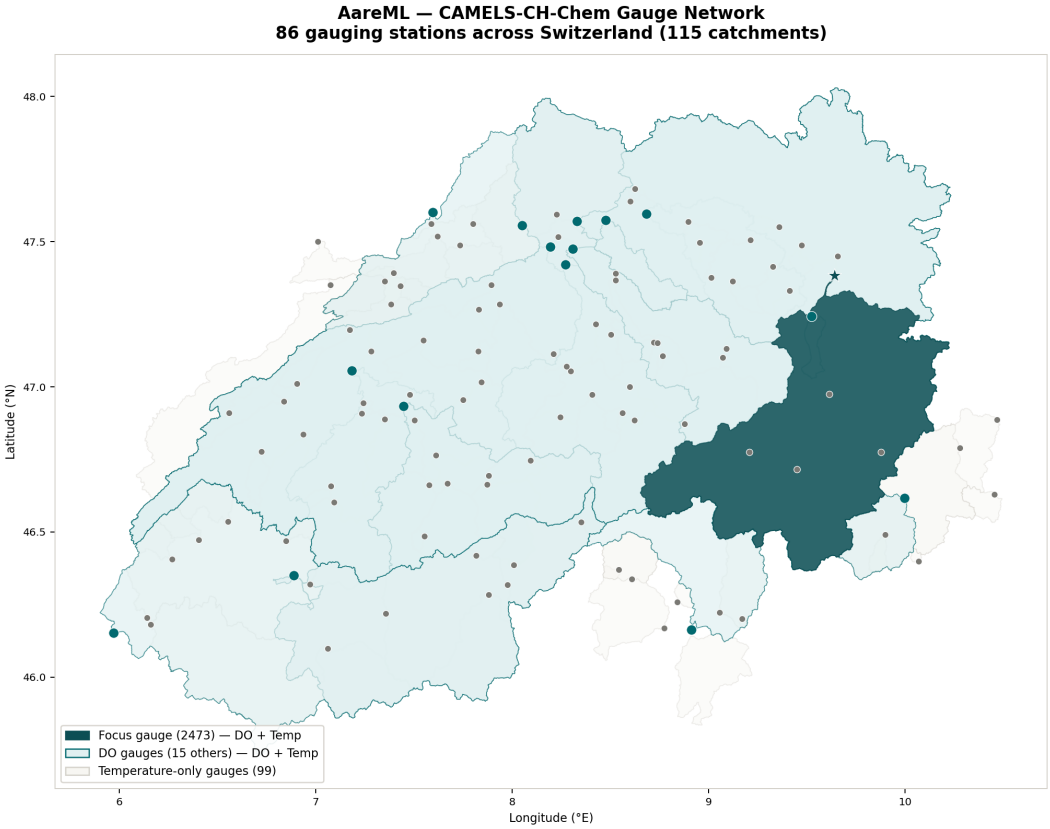

Figure A0: CAMELS-CH-Chem gauge network across Switzerland. Dark teal = focus gauge 2473 (catchment highlighted). Teal = 15 other DO-capable gauges. Light = 70 temperature-only gauges. Catchment boundaries shown for all 86 stations.

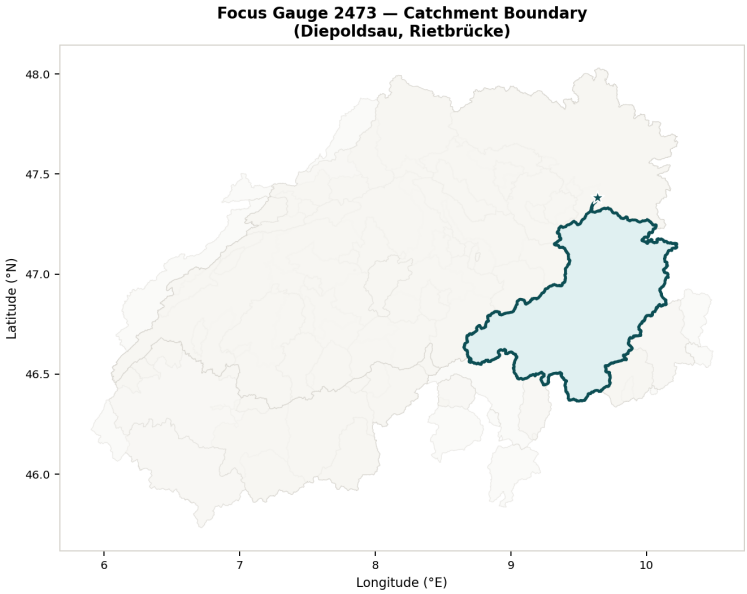

Figure A0b: Focus gauge 2473 catchment boundary (dark teal) shown against all CAMELS-CH-Chem catchments (light).

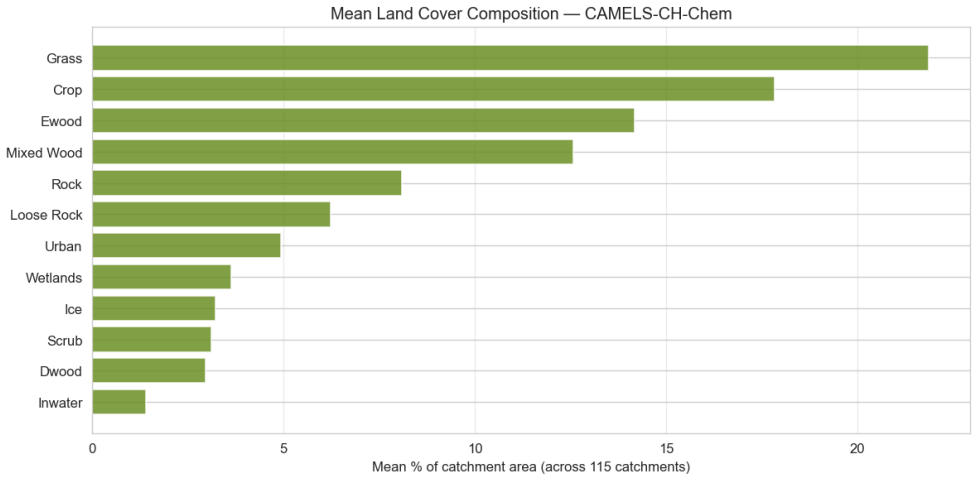

Figure A1: Land cover composition by catchment, ordered by DO gauge ID. Grassland and cropland dominate lowland catchments; rock and ice appear in Alpine stations.

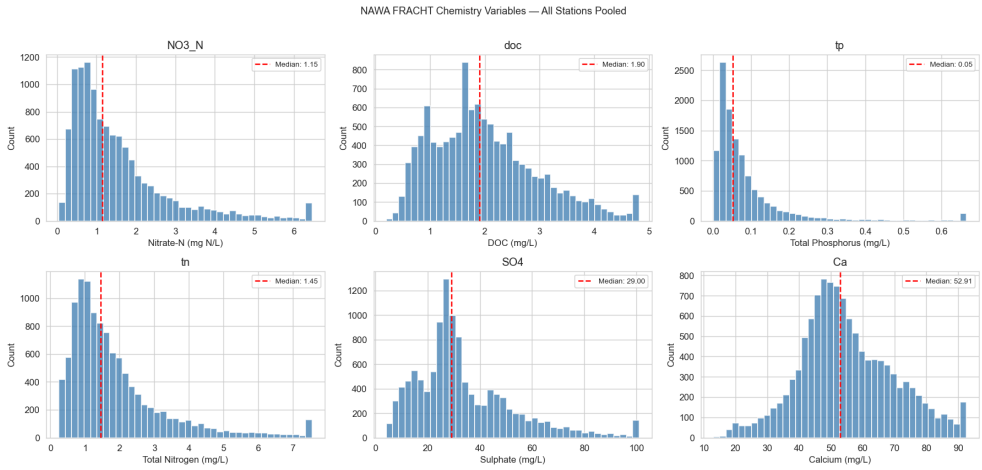

Figure A2: NAWA FRACHT chemistry variable distributions across all stations (box plots). Heavy-metal concentrations show strong right skew.

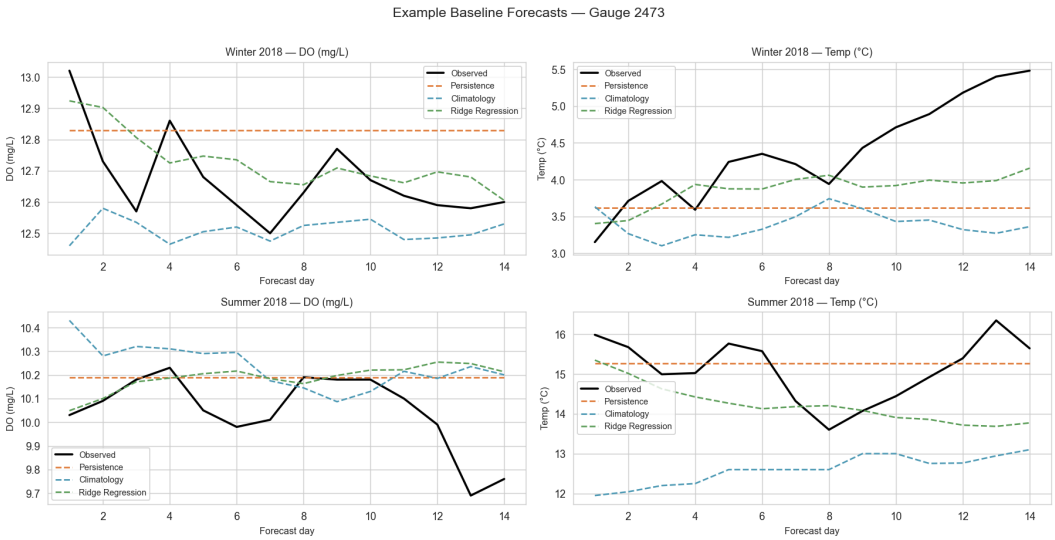

Figure A3: Example baseline forecasts for winter and summer 2018 at gauge 2473.

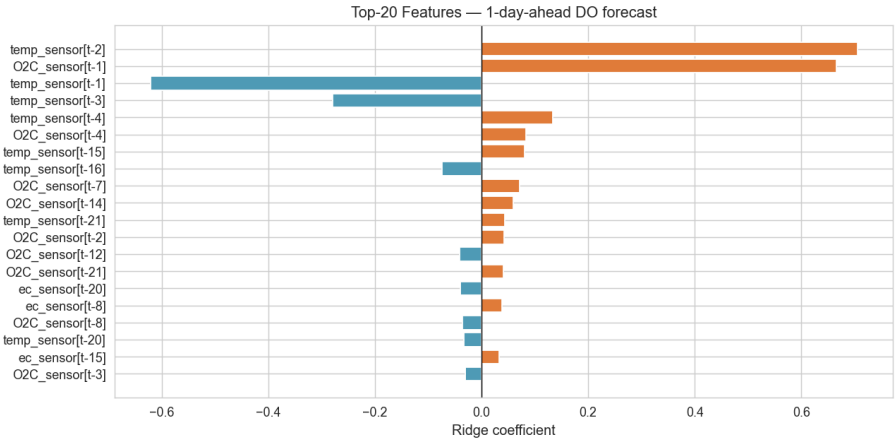

Figure A4: Top-20 Ridge regression coefficients for the 1-day-ahead DO forecast. Recent DO lags dominate, as expected.

B Repository Structure

The full source code, notebooks, and results are publicly available on GitHub: github.com/polar-bear-after-lunch/AareML

```
AareML/
notebooks/
01_data_exploration.ipynb
02_baselines.ipynb
03_lstm_single_site.ipynb
04_multisite_analysis.ipynb
04b_multisite_temperature.ipynb
05_shap_interpretation.ipynb
06_cross_ecosystem_lake.ipynb
07_lake_eda.ipynb
08_usgs_transfer.ipynb
src/
config.py data.py metrics.py model.py impute.py
figures/ results/ data/ (excluded from git)
download_data.py README.md requirements.txt .gitignore
```

C Reproducibility

All random seeds are fixed to 42 throughout. The full codebase is at github.com/polar-bear-after-lunch/AareML. Datasets are publicly available: CAMELS-CH-Chem at zenodo.org/records/14980027 and LakeBeD-US at huggingface.co/datasets/eco-kgml/LakeBeD-US-CSE. Run python download_data.py to fetch all data automatically, then run notebooks 01–07 in order.

Appendix E — Glossary

Key terms used throughout this report, grouped by domain.

Term	Definition
Hydrology & Water Quality	
Gauge / Gauging station	A fixed measurement point on a river where water level, flow rate, and sensor readings (temperature, pH, dissolved oxygen, etc.) are continuously recorded. CAMELS-CH-Chem has 86 gauging stations across Switzerland.
Catchment / Watershed	The land area that drains into a specific river or gauging station. Rain falling anywhere in a catchment eventually flows to the gauge. Catchment size, elevation, and land cover strongly influence water quality.
Dissolved oxygen (DO)	The amount of oxygen dissolved in water, measured in mg/L. Critical for aquatic life — below 5 mg/L stresses fish populations; below 2 mg/L causes mass mortality. DO is the primary prediction target in AareML.
Electrical conductivity (EC)	A measure of water's ability to conduct electricity, reflecting dissolved ion concentration (µS/cm). Higher EC often indicates agricultural runoff or urban influence.
pH	A measure of acidity/alkalinity on a scale of 0–14. Natural Swiss rivers typically range from 7–8 (slightly alkaline).
NAWA FRACHT	Swiss federal river chemistry monitoring programme (Nationale Beobachtung Oberflächengewässer — loads). Provides monthly grab samples of nutrients (NO3, NH4, TP, TN), dissolved organic carbon (DOC), and discharge.
Reaeration	The natural process by which oxygen enters river water from the atmosphere. Rivers have higher reaeration rates than lakes because turbulent flow continually exposes water to air — a key reason river DO is more predictable than lake DO.
Stratification	In lakes, the separation of water into distinct temperature layers that resist mixing. Summer stratification cuts off deep water from atmospheric oxygen, causing DO depletion — a major source of unpredictability not present in rivers.
NSE	Nash-Sutcliffe Efficiency — a metric for model skill. NSE = 1 is perfect; NSE = 0 means the model is no better than predicting the mean; NSE < 0 means the model is worse than the mean.
KGE	Kling-Gupta Efficiency — decomposes model error into correlation, bias, and variability components. KGE = 1 is perfect. Often preferred over NSE for hydrological time series because it equally weights all three error components.
RMSE	Root Mean Squared Error — the square root of the average squared difference between predictions and observations. Lower is better. Reported in the same units as the target (mg/L for DO, °C for temperature).
Machine Learning	
Lookback window	The number of past days of sensor data fed into the model as input. AareML uses a 21-day lookback — the model sees the last 3 weeks of temperature, pH, EC, and DO before making a forecast.
Forecast horizon	How many days ahead the model predicts. AareML uses a 14-day horizon, matching the LakeBeD-US benchmark.

Seq2Seq LSTM	Sequence-to-sequence Long Short-Term Memory network. An encoder reads the input sequence (lookback window) and compresses it into a hidden state; a decoder then generates the output sequence (14-day forecast) step by step.
Teacher forcing	A training technique where the decoder receives the true target value as input at each step (instead of its own prediction). This stabilises early training. AareML linearly decays teacher forcing from 50% to 0% over the first half of training.
Optuna / TPE	Optuna is a hyperparameter optimisation library. TPE (Tree-structured Parzen Estimator) is its default algorithm — it builds a probabilistic model of which hyperparameters perform well and samples promising candidates.
SHAP	SHapley Additive exPlanations — a method to explain model predictions by assigning each input feature a contribution score. GradientSHAP (used here) computes these scores using gradient information from the neural network.
Zero-shot transfer	Applying a model trained on one site directly to a new site without any retraining. In AareML, the LSTM trained on gauge 2473 is evaluated on 11 other gauges in zero-shot mode.
Block bootstrap	A method to estimate confidence intervals for time-series metrics. Instead of resampling individual days (which would break temporal structure), contiguous blocks of days (block size = 30) are resampled. AareML uses 500 bootstrap replicates.
Entity-Aware LSTM (EA-LSTM)	A variant of LSTM where static catchment attributes (area, elevation, land cover) modulate the input and cell gates, allowing a single model to adapt to multiple catchments simultaneously. Based on Kratzert et al. (2019).
Early stopping	A regularisation technique that stops training when validation loss stops improving for a set number of epochs (patience). Prevents overfitting. AareML uses patience = 12 epochs for the final model.
Dataset & Evaluation	
CAMELS-CH-Chem	Catchment Attributes and Meteorology for Large-sample Studies — Switzerland — Chemistry. A dataset of daily sensor measurements and monthly chemistry samples at 86 Swiss river gauges (1981–2020). Published by Nascimento et al. (2025).
LakeBeD-US	Lake Benchmark Dataset — United States. A standardised benchmark for lake water quality prediction covering 21 US lakes. The LSTM benchmark (DO RMSE = 1.40 mg/L) is the primary reference point for AareML. Published by McAfee et al. (2025).
Train / Val / Test split	Chronological division of data into training (≤2014), validation (2015–2016), and test (2017–2020) periods. Chronological splitting is essential for time series — random splitting would leak future information.
Imputation	Filling in missing values. AareML uses training-set column means to fill NaN values in input windows. The more sophisticated SAITS method (self-attention based) is implemented in src/impute.py for future use.

Appendix D — Report Version History

This report is a living document updated iteratively as computational results become available. The table below records each version, its date, and the key changes made.

Version	Date	Key Changes
1.10	April 2026	Cross-continental zero-shot transfer to 4 US rivers (notebook 08); statistical significance confirmed for zero-shot LSTM vs Ridge (p=0.005); NSE+MSE combined loss implemented.
1.9	April 2026	Updated single-site and multi-site results from v1.20 hyperparameter run (75 Optuna trials, ReduceLROnPlateau, 3-seed ensemble). LSTM (default) DO RMSE = 0.308 mg/L (KGE = 0.855); LSTM (best) DO RMSE = 0.319 mg/L (KGE = 0.942). Multi-site transfer: 0.427 mg/L zero-shot (12 gauges), 0.388 mg/L per-gauge (11 gauges), 0.417 mg/L EA-LSTM. Corrected gauge 2068 temperature RMSE from 3.16°C to 1.36°C. Added multisite gauge map and RMSE bar chart (Section 5.3).
1.8	April 2026	Real temperature multi-site results (15 gauges, mean RMSE 2.59°C, NSE=0.730); EA-LSTM added to multi-site DO comparison (mean RMSE 0.406 mg/L); ReduceLROnPlateau + 3-seed ensemble added; Section 5.3b fully populated with per-gauge table, elevation stratification analysis, and training/distribution/correlation figures. Abstract updated with temperature transfer summary. Teal change-highlight markers added to new/updated sections.
1.6	20 Apr 2026	Added Switzerland gauge + catchment map (Appendix A). Added notebook 04b (temperature multi-site; 15 gauges evaluated in v1.8). Added glossary (Appendix E, 25 terms). Extended Section 6.1 with three-mechanism discussion of river-lake gap. Added temperature critical thresholds to Section 1. Report version and timestamp on cover page. 3-model bug re-audit (GPT-5.4, Opus 4.7, Sonnet 4.6): 11 additional bugs fixed (v1.13–1.16). Test suite added: 53 pytest tests, all passing.
1.5	15 Apr 2026	Added cross-ecosystem results (notebook 06, Lake Mendota / LakeBeD-US): Ridge baseline on lake achieves DO RMSE = 1.030 mg/L (3.4× higher than river). AareML Ridge outperforms the published LakeBeD-US LSTM (1.40 mg/L). Section 5.5 added. Notebook 07 (Lake Mendota EDA) added.
1.4	15 Apr 2026	Added SHAP attribution results (notebook 05, GradientSHAP, 500 windows): temperature[t−1] is the dominant driver (mean |SHAP|=0.644), ahead of DO[t−1] (0.527). Effective LSTM memory = 3–4 days despite 21-day lookback. Section 5.4 added.
1.3	14 Apr 2026	Multi-site transfer results (notebook 04, 12 gauges): zero-shot DO RMSE = 0.445 ± 0.083 mg/L (3.1× better than LakeBeD-US), per-gauge DO RMSE = 0.396 ± 0.092 mg/L (3.5× better). Latitude/northing: strongest catchment predictor (Spearman ρ = 0.78). Section 5.3 updated.

Version	Date	Key Changes
1.2	12 Apr 2026	LSTM single-site results (notebook 03, 20 Optuna trials): LSTM (default) DO RMSE = 0.299 mg/L (KGE = 0.918), LSTM (best, hidden=256) DO RMSE = 0.301 mg/L (KGE = 0.927). All models 4.7× better than LakeBeD-US LSTM. 30 peer-review fixes applied.
1.1	07 Apr 2026	EDA results (notebook 01): data availability, time series, seasonal cycle, land cover. Baseline results (notebook 02): Ridge DO RMSE = 0.303 mg/L, NSE = 0.888, KGE = 0.908 with block-bootstrap 95% CIs.
1.0	06 Apr 2026	Initial draft: project proposal, dataset description (CAMELS-CH-Chem, 86 gauges), methods outline, related work.

This report will be finalised for submission by 15 June 2026. Remaining work includes the catchment-attribute SHAP surrogate model (requires running notebooks 04 and 05 sequentially on UBELIX HPC), final polish of discussion sections, and proofreading.